# ESPORTS: CSTCG

## MASTER GAME DESIGN DOCUMENT

COLLECT · COACH · SIMULATE · TRADE




DUST II 3D SPATIAL TEST BENCH · PROTOTYPE MODEL VIEW

**DESIGN BASELINE / VERSION 2.0 / 2 NOVEMBER 2026**

PC + MOBILE | GODOT 4.x | SERVER-AUTHORITATIVE TARGET

Consolidated from project design discussions and Godot prototypes through spatial build v1.4.

# DOCUMENT CONTROL

**Scope:** Definitive design intentions, numerical rules, current prototype status, engineering contracts, balancing assumptions, and outstanding questions.

**Authority hierarchy:** Later explicit decisions supersede earlier examples or provisional suggestions. An item marked PROTOTYPE is not automatically an approved final production rule.

**Current critical dependency:** The 3D Dust II world exists as a separate geometry/test scene; the fully featured legacy match engine has not yet been migrated into it.

## SECTION MAP

| Sections | Contents |
|---|---|
| 01–04 | Product vision, platforms, core loop, onboarding and Pubstompers |
| 05–11 | Card identities, attributes, rarity, promotions, mods, behavior and chemistry |
| 12–15 | Roles, coaching, strategies, formats and match-state engine |
| 16–23 | 3D Dust II, control of space, tactical AI, combat, armory, grenades, economy and objectives |
| 24–30 | Spectator experience, analytics, packs, events, marketplace, servers and data model |
| 31–35 | Quality assurance, prototype history, roadmap, risks and decision register |
| App. A–C | Illustrative match, acceptance-test matrix and glossary |

The Word document uses hierarchical headings for navigation. Use **View → Navigation Pane** in Word to jump directly to chapters.

**How to read decision status**

| Label | Meaning |
|---|---|
| **LOCKED** | Explicit design decision accepted by the creator. |
| **TARGET** | Intended feature/behavior reaffirmed during prototype development, but may need implementation and tuning. |
| **PROTOTYPE** | Exists in one or more Godot experimental builds; engine-level correctness and full integration are not necessarily verified. |
| **PROPOSED** | Recommended direction or sample tuning, not creator-approved as a permanent rule. |
| **OPEN** | Requires a future decision, measurement, licensing check, or test result. |

## 01 — Executive Vision

**ESPORTS: CSTCG** is a cross-platform, free-to-play competitive Counter-Strike esports card-collection and team-management simulation. Customers acquire unique player-card instances, develop them with duplicate promotions, permanent modifications, and temporary training, build rosters with hidden interpersonal dynamics, coach through a tactical philosophy, and compete in genuine event-driven 5v5 matches. It is **not** a card-on-card battle game and not a first-person shooter controlled by the customer.

A match is generated from ten simulated people acting under uncertain information in a spatial world: they buy weapons, use utility, take and deny space, trade, plant, retake, save, and adapt. A winner is **never** chosen in advance for the event stream to justify. The simulation is the authority for every statistic and every attributable contribution.

**Experience promise.** A lower-rarity roster with coherent tactics, chemistry, economy, and useful roles can defeat a lineup of numerically exceptional players. Conversely, exceptional individual cards remain exciting because distinct distributions, ability ceilings, sniping profiles, and individual agency create varied styles.

### Design pillars

1. **Unique collectibles, not interchangeable portraits.** Cards for the same professional can differ in rarity, natural stats, hidden behavior, modifications, history, and edition.
2. **Real teamwork rather than an OVR shortcut.** Trading, communication, utility, roles, controlled territory, and decision timing matter.
3. **Causal, auditable simulated Counter-Strike.** Replayable rounds, credible geometry, actual equipment and economy, and attribution tied to events.
4. **Long-term roster attachment.** Chemistry grows, substitutions disrupt teams, Pubstompers are personally named, and competitive records follow exact card IDs.
5. **Collecting and trading with supply integrity.** Sealed tradable packs, events, reissues, provenance, public drop odds, and a marketplace.
6. **Accessible automation with optional depth.** Automatic Coaching is default; expert players can intervene at authorized moments.

### Product boundaries

**LOCKED:** PC and mobile; five active players; mostly customer-vs-customer; bot scrims/tournaments; real-time queues and competitions; opt-in asynchronous competition; free-to-play with purchasable credits; realistic MR12/MR8 match simulation; top-down tactical spectator and killfeed; backend-only Alpha/Bravo/Charlie; rarity and tier independent; no duplicate professional identities on one active five; original/implied tournament organizers rather than real-world TO brands.

**OPEN:** Final commercial title, exact game ratings/age classification, launch regions, approved licenses and data feeds, release business model safeguards, ranked rules, server budget, market fraud protection, and final numerical balance.

## 02 — Audience, Platforms & Session Model

**Intended audience.** Counter-Strike and esports followers; trading-card collectors; fantasy-team and sports management players; tactical theorycrafting enthusiasts; people who enjoy watching AI-simulated outcomes and explaining why they occurred. Management and watching are the product; first-person weapon control is not a required mechanic.

**Platforms — LOCKED:** Native PC and mobile clients with the same authoritative game rules. Responsive UI may differ; card IDs, marketplace, eligibility, and outcomes must stay consistent across platforms. Cross-play and synchronized inventory are expected from one backend identity, subject to platform compliance.

**Opponent model — LOCKED:** Primary competition is other customers. Real-time matchmaking queues and live tournaments pair active customers. Bot scrims and tournaments provide practice. Customers can explicitly opt their roster into **offline engagement** for asynchronous scrims and tournaments; passive roster use without opt-in is not permitted.

**Session pathways — TARGET:** Brief collection management, accelerated simulated matches, live radar viewing with manual calls, tournament participation, and post-match analysis. Live, accelerated, and instant presentations must share the same authoritative match events; speed choice alone must not re-roll results. Manual coaching introduces deliberate pause points and therefore may require a separate real-time match procedure.

**No forced fixed play pattern.** Customers can focus on assembling competitive teams, trading and opening collections, developing a named Pubstomper, or watching tournaments. The game needs an intelligible first-session path without requiring knowledge of grenade lineups or real-world IGL styles.

## 03 — Core Loop & Meta-Progression

**Collect → inspect → train → configure → simulate → analyze → earn → upgrade → trade → repeat.** New card pulls and marketplace transactions change the available lineup; training provides short-term boosts; promotions increase modification capacity; chemistry rewards established partnerships; match statistics and observed interactions inform future decisions.

**A normal management cycle:** review five selected player cards and their equipment-independent stats; pick side-specific roles and strategies; adjust the Playbook–Midrounding slider; select Automatic or Manual Coaching; enter a live queue or eligible offline/bot event; watch the killfeed/radar or speed through; study impact and tactical errors; receive credit and progression rewards; decide whether to retain, sell, modify, promote, or train cards.

**Currencies.** One named **Credits** currency is locked for purchasing packs and market assets, and is earnable through play. Credits may also be purchased with real money. No second premium currency is confirmed. Do not confuse persistent account Credits with the **round economy dollars** that the sim spends on CS equipment.

**Progression tracks — OPEN:** A manager/coach/team level (name unresolved) unlocks a subset of nontradable strategy cards, whereas eligible alternative strategies can trade. Exact XP, level curve, prestige, rewards, match eligibility, and season-reset behavior remain to be designed.

## 04 — Starting Experience & Pubstompers

**Starter roster — LOCKED:** Every new customer begins with **four Pubstomper cards**. A Pubstomper is a faceless customizable player whom the owner may name and assign roles. Customers also select **one professional from twenty Common T5 options**, forming an initial five. A **starter gift pack** includes one guaranteed Peak professional and four additional generated cards. A starter-only eligible pool is being considered but is not yet finalized; do not assert that its contents or odds are fixed.

**Pubstomper acquisition — LOCKED:** Further Pubstomper pulls can be used as development material on an already named Pubstomper or converted into an additional customizable character. Each created Pubstomper requires a unique character identity, independent behavior roll, persistent card ID, and ordinary roster legality rules. Their presentation is anonymous/customizable, avoiding forced association with a real pro.

**Pubstomper promotion — LOCKED:** The intended duplicate/card requirement is **10×** the equivalent ordinary promotion cost. With the established tier costs, this implies 100 points (T5→T4), 250 (T4→T3), 400 (T3→T2), and 550 (T2→T1), for 1,300 cumulative points **if** Pubstomper promotion uses the same points denomination. How individual Pubstomper pull values map to those points is **OPEN**, so these figures are a mathematically consistent interpretation, not proof of an approved redemption schedule.

**Starter onboarding — TARGET:** Name the four Pubstompers; explain basic roles; choose the Common T5 professional; reveal and open the starter gift; preview a five-player lineup; simulate a guided bot round; show why economy, positioning, trading, and teammate choice matter; introduce the card inspector and management screens. Avoid forcing marketplace participation in the tutorial.

**Abuse protection — PROPOSED:** Unique IDs for each created Pubstomper, rename history, marketplace identity disclosure, prevention of duplicate use of a single character within a lineup, and clear restrictions on offensive or misleading names.

## 05 — Card Taxonomy, Identity & Ownership

| Asset class | Core function | Transfer policy |
|---|---|---|
| **Professional player card** | Individually rolled natural stats, rarity, tier, hidden behavior, history and provenance. | Tradable, including modified/promoted instances — LOCKED. |
| **Pubstomper card** | Named, faceless custom character or promotion material. | Exact trading and identity-transfer details OPEN. |
| **Modification card** | Permanent stat adjustments on an individual player-card slot. | Tradable before application; fate after replacement OPEN. |
| **Strategy card** | Enables/weights tactical plan or coaching behavior. | Tradable subset; progression-locked subset — LOCKED. |
| **Sealed pack** | Immutable collectible container with collection and drop-table provenance. | Tradable — LOCKED. |
| **Event/challenge reward** | Special-edition cards/packs earned through active content. | Transferability per reward rule — OPEN. |
| **Future Artifact eligible category** | Possible new collectible type for reissue packs. | OPEN; must be versioned. |

**Identity constraints — LOCKED:** The customer can hold multiple independently developed cards depicting the same professional. A single active five-player roster can contain **only one instance of a given professional**. Professional identity, player-card instance ID, card edition, acquisition method, and collection membership are distinct database properties. Two similarly named or similarly rated cards are not the same collectible.

**Card layers:** Immutable rolled **base statistics**; immutable historical template/identity; fixed secret behavioral archetype; **tier** and unlocked slots; permanently applied mod effects; temporary training effects; match-history and observation state; edition/foil/provenance and transfer history. A market inspector should display rolled base numbers and modified effective numbers separately; temporary buffs should never be advertised as permanent.

**Versioning.** Match logs record exact player-card IDs, stat/effect versions, team and manager identity snapshots, content patch version, and seed so later balance changes cannot retroactively rewrite what occurred.

# 06 — Player Attributes & Hidden Sniper Profile

## 06.1 Eight main visible attributes — LOCKED

| Attribute | Match interpretation |
|---|---|
| **Aim** | Mechanical shot placement, firing control and hit quality, not engagement selection. |
| **Reaction** | Detection-to-response timing and first-contact latency. |
| **Positioning** | Cover selection, spacing, peek choice, exposure, routes and leverage. |
| **Utility** | Grenade choice, lineup fidelity, timing and teammate coordination. |
| **Gamesense** | Interpretation of incomplete evidence, predictions, saves, reads and adaptation. |
| **Clutch** | Decision-making and performance in genuinely pressured numerical situations. |
| **Teamwork** | Trading, communication, support, timing, synchronization and cooperation. |
| **Consistency** | Width of form/performance distribution; not an automatic average-performance debuff. |

**Scales — LOCKED:** Natural statistics typically occupy roughly 0–100, but there is **no universal hard cap**. Exceptional rolls or profiles can exceed 100, and a specific professional may be unable to reach 100 in some areas even at Peak. Applied modifications may exceed natural limits. All detailed per-stat conversion curves remain provisional.

**Generation — LOCKED:** Combine permitted historical esports performance evidence with designer judgment. Opening duels, ADR, clutch outcomes and utility events are useful data proxies; abstract properties such as Gamesense, Positioning and Teamwork cannot be directly inferred from a single raw dataset. Templates can be career-wide for ordinary cards and event/era-specific for themed cards.

**Individual variability — LOCKED:** Same pro/same rarity cards normally deviate by **no more than about 10 points in either direction on average**, but this is not a universal limit. Draws persist and are never rerolled on tier promotion.

## 06.2 Inspect-only advanced sniping — TARGET / PROTOTYPE

| Hidden-on-face inspection stat | Use in the engine |
|---|---|
| **Passive Sniping** | Angle holding, patience, stability, defensive AWP/Scout pressure. |
| **Active Sniping** | Proactive re-peeks, rotations, finding opening angles and repositioning. |
| **Combat Sniping** | Pressure performance, close-angle sniper fights and recovery after contact. |

These three statistics **do not appear on the card face**, but are disclosed in the card inspector. Distinguish them from Alpha/Bravo/Charlie, which must never be explicitly exposed to the customer. Sniping stats are unique per card instance, with exact generation correlations and modifier eligibility OPEN.

### 06.3 Variance and simulation use — TARGET

An agent should first find or be given an opportunity; its Aim and Reaction should determine how it executes that opportunity. A high-Aim player with weak Positioning may frequently take poor duels, while a moderate-Aim player with excellent positioning can create favorable fights. Low Consistency raises volatility: a spectacular performance and a disastrous one should both remain possible. Player roles, weapon handling, health, utility, map position and delivered information all modify individual decisions and hit probabilities.

## 07 — Rarity, Ceilings & Card Generation

**Rarity ordering** — **LOCKED: Floor → Common → Uncommon → Semi-Rare → Rare → Legendary → Peak → Divine**. The highest rarity is spelled **Divine**. Floor is deliberately almost comically weak. Rarity defines possible stat ceilings, distribution shapes, extremity, and acquisition odds; it is distinct from promotion tier.

A high rarity does **not** guarantee a conventional well-rounded card. Some Peak or Divine cards may have exceptional strengths and serious weaknesses; others may have almost no weaknesses. A Legendary may outperform an unusually extreme Peak for a specific lineup. Likewise, a card's total attributes are not identical to its tactical or market value.

**Peak drop probability** — **LOCKED range:** Pack-dependent chance from **0.50% down to 0.001%**; the exact probability table by pack has not been approved. No separate numerical Divine probability is confirmed. Do not assume equal odds among rarities, uniform source-pool assignment, or a guaranteed high-tier starter outside the specially guaranteed Peak starter gift.

**Generation pipeline** — **TARGET:** (1) select pack version and rarity through published odds; (2) select eligible professional/template by pack-specific weights; (3) sample profile from rarity-conditioned range under professional ceilings; (4) apply per-instance stat variation; (5) sample one permanent hidden behavioral type and inspect-only sniper stats; (6) instantiate T5 card and immutable provenance record; (7) commit an auditable server transaction. Event-era versions may differ from career templates.

**Sample balancing only:** A high-rarity card might have Positioning 110 and Consistency 30, while a different roll has no meaningful downside. Such illustrations are not factual ratings of any named professional.

## 08 — Tier Promotion, Duplicate Costs & Slots

**Tier ladder** — **LOCKED:** All professional cards begin at **T5**, and can promote through T4, T3 and T2 to T1. **Promotion never changes rolled base stats.** It unlocks exactly **two additional modification slots belonging to the newly attained tier**, while preserving older slots and existing mods.

| Promotion step | Points needed for that step | Cumulative points | Available slots after step |
|---|---|---|---|
| Initial **T5** | — | 0 | 2 × T5 |
| **T5 → T4** | 10 | 10 | 2 × T5 + 2 × T4 |
| **T4 → T3** | 25 | 35 | + 2 × T3 (6 total) |
| **T3 → T2** | 40 | 75 | + 2 × T2 (8 total) |
| **T2 → T1** | 55 | 130 | + 2 × T1 (10 total) |

**Promotion material** — **LOCKED:** Sacrifice additional **cards of the same professional identity** as the target, valued by the rarity of the sacrificed card. The developed target is never consumed. Any duplicate may be

sacrificed, **including cards with modifications or previous promotions**. Those previous investments do not automatically increase its rarity-derived value.

| Consumed rarity | Promotion points |
|---|---|
| Floor | 1 |
| Common | 1 |
| Uncommon | 2 |
| Semi-Rare | 3 |
| Rare | 3 |
| Legendary | 4 |
| Peak | 5 |
| Divine | 6 |

**Overspend — LOCKED:** Excess points **do not roll forward** to another promotion. Convert the excess to a **very small** Credits refund to soften the loss; the refund exchange rate is OPEN. Example: spending two Divine copies (12 points) on a 10-point step gives the buyer 2 overage points' worth of a small credit refund, not 2 points stored for T3.

**Slot compatibility — LOCKED:** A mod can occupy a matching-tier or **higher-tier** available slot (a T5 mod can use a T3 slot); a higher-tier mod cannot occupy a lower-tier slot. This flexibility should be surfaced explicitly to avoid wasting valuable slots. Stacking and replacement-destruction rules remain OPEN.

## 09 — Permanent Mods, Temporary Training & Cooldowns

**Permanent modifications — LOCKED:** Mods modify card-instance effective statistics (positively and/or with tradeoffs) without changing rolled base values. Applying any mod starts a cooldown on that **player-card instance**, not on the mod tier. Cards remain eligible to play and trade while cooling down; how transferred cooldown state behaves is OPEN.

**Modification cooldown — LOCKED: 10 eligible completed matches OR 10 real-world hours, whichever occurs first.** The clock and match-counter start when the mod is applied. Eligibility of casual/offline matches for reducing the counter needs a definitive anti-abuse rule.

**Replacement — LOCKED:** Mods cannot simply be removed to recover the original slot state; **replacing** an equipped mod is allowed. Whether the replaced mod returns to inventory, is destroyed, or is converted is OPEN. A later mod attempt still respects the card-level cooldown unless another explicit exception is approved.

**Training — LOCKED:** Temporary training increases effective stats for **3 eligible matches OR 2 real-world hours, whichever occurs first**. Training is independent of permanent mods and does not alter base stats. Suggested drills include Aim, Demo Review, Utility Lineups, Deathmatch, Scrims, Clutch Practice, and Positioning; exact drill formulas, stacking and availability are OPEN.

**Inspector requirement:** Display *rolled base / applied modification delta / active training delta / final effective* values separately, with expiration/cooldown timers. Marketplace views must not disguise expiring bonuses as an inheritable permanent stat.

## 10 — Hidden Behavioral Archetypes

**Alpha, Bravo and Charlie — LOCKED backend-only types.** Each individual card instance permanently receives one of the three, including different cards depicting the same professional. These are game-simulation dispositions, **not public labels, personality diagnoses, or objective descriptions of real people**.

**Core interaction tendencies — LOCKED qualitatively:** One Alpha with two to four compatible Bravo or Charlie personalities often creates a viable leadership shape; multiple Alpha players may clash; mixed Bravo/Charlie groups can have friction; an all-Bravo or all-Charlie roster can attain excellent chemistry, while exhibiting positional or adaptability weaknesses. Specific archetype-to-archetype coefficients are not yet locked.

**Never show the A/B/C labels.** Customers discover interaction patterns through on-screen behavior, match outcomes, player inspector observations, team reports and optional team journal: conflicting rotation calls, coordinated trades, reliable support, improvised play, hesitations, or repeated spacing problems. Observations should be linked to actual causal events; they must not be randomly fabricated flavor text.

**No simplistic deciphering:** A tactical disagreement is not definitive evidence of Alpha status. It could also reflect insufficient information, a poorly fitting strategy, low chemistry, low Gamesense, or a particular situation. Behavioral type influences choice probabilities and interaction patterns; it is not a hard script that makes every Alpha disobey or every Bravo obey.

## 11 — Chemistry: Growth, Ceilings & Substitutions

**Chemistry begins at zero** for a genuinely new roster and improves through eligible shared experience, approaching a **hidden maximum determined solely by the five cards' Alpha/Bravo/Charlie composition**. Player attributes, team roles and previous relationships can affect moment-to-moment coordination, but **must not change the compatibility ceiling** under the currently locked rule.

The displayed customer value can be a number/trend, and contextual observations can communicate the trajectory. **Never reveal the exact ceiling or backend archetypes.** Teams with equal displayed overall chemistry can differ in which tactical interactions feel synchronized or awkward, but any proposed hidden secondary dimensions should be treated as implementation detail rather than extra factors modifying the agreed ceiling.

**Growth — TARGET:** Gradually converge on ceiling with diminishing returns. Example only: $C_next = C + g \times (C_limit - C)$, with $g$ and rounding/calibration OPEN. A bad composition may approach a poor ceiling; a full Bravo/Charlie roster may approach an excellent ceiling yet still struggle with positioning.

### 11.1 Roster-change depreciation — LOCKED

Let $C_0$ be **the chemistry immediately before a group of substitutions**, not the old roster's lifetime maximum. On swapping cards, apply an immediate penalty; over the next **3 eligible matches**, chemistry depreciates toward an **offset from $C_0$**. It then adjusts toward the new lineup's hidden ceiling, including further decline if that ceiling is lower. Negative chemistry is possible and represents active incompatibility, not merely lack of experience.

| Number of replacements | Immediate value | Three-match depreciation target |
|---|---|---|
| **1** | $C_0 - 10$ | $C_0 - 30$ |
| **2** | $C_0 - 10$ | $C_0 - 40$ |
| **3** | $C_0 - 15$ | $C_0 - 40$ |
| **4 or 5** | Reset to 0 | Follow-on adjustment behavior OPEN |

**Example:** An 85-chemistry roster replaces one card: 85 → 75 immediately → toward 55 during three adjustment matches → toward the replacement lineup's hidden ceiling thereafter. If the new ceiling is 40, chemistry keeps moving down toward 40 instead of rebounding. A roster at chemistry 20 can fall to −10 by the end of a one-card adjustment.

**Reverting to a historical roster — LOCKED qualitative behavior:** Recreating a previous lineup still causes disruption, **with higher highs and lower lows / increased volatility** compared with a totally unfamiliar group. Exact historical familiarity memory, initial conditions, and variance multipliers are OPEN. Do not simply restore prior chemistry with no disruption.

**Design guardrails:** Specify whether several edits before the next match count as one group; how a substitute is counted when later reversed; how negative chemistry is clamped, if at all; whether scrims progress chemistry (the prototype currently excludes scrims); and how elapsed time interacts with a 3-match depreciation period. These are still engineering/product decisions.

## 12 — Team Construction, Roles & Coaching Philosophy

**Rosters — LOCKED:** Exactly five active cards, distinct professional identities, valid match availability and side rules. Assignable roles can include Entry, Support, IGL, AWPer, Lurker, Anchor and Rotator. T-side and CT-side roles can be distinct; **automatic assignment is default**, while expert manual role specification is allowed and may either improve or harm a strategy.

**Coaching mode — LOCKED: Automatic Coaching is the default; Manual Coaching is optional.** Both use the exact same match simulation, without arbitrary bonuses. Auto selects buys and decisions within the prepared roster, playbook and IGL tendencies; manual mode gives the customer authorized interventions but does not directly control individual shots or fabricate outcomes.

**Playbook vs. Midrounding slider — LOCKED:** The roster screen has a continuous bias control. Left favors rehearsed executes, roles and structure; right favors improvisation, dynamic calls, and mid-round agency. Player and IGL effectiveness changes with their preferred style. Using AleksiB, karrigan and apEX as **proposed card-style examples**: playbook-heavy, freedom-heavy and hybrid, respectively; these are gameplay design examples, not verified claims about real individuals.

**Agent autonomy — LOCKED: B** is the default: players generally follow leadership but make independent decisions. The slider modulates team autonomy toward stricter execution or freer decisions. Particular cards may be extremely obedient or frequently disregard calls regardless of the slider. This should emerge from decision mechanics, not an unexplained flat attribute bonus.

**Manual calls — LOCKED: MR12:** three tactical pauses per team and additional coach calls at Round 1 and Round 13; **MR8:** one tactical pause and no automatic half-start call. The precise timeout duration, who can invoke it, queue synchronization and what changes are legal during a pause remain OPEN. The original between-round-only concept was superseded by the explicit timeout/half-call design.

## 13 — Strategies, Playbooks & Unlocks

**Acquisition — LOCKED:** A subset of tactics is earned freely through manager/coach/team progression and is **not market-tradable**. Other strategy cards can be obtained from eligible drops, events or the marketplace. Exact progression title, unlock tables and duplication mechanics are OPEN.

**Examples of attacking patterns:** Default, A Long take, Mid/Cat split, A Short execute, Mid-to-B split, B tunnels execute, contact walk, rush, fake, slower map control, late rotations, and committed post-plant setups. **Examples of CT patterns:** passive holds, Mid Doors pressure, Cat containment, site stacks, information pushes, defensive utility, layered crossfires, coordinated retakes, and economy saves.

**Strategies are policies, not percentage boosts.** An A Short execute instructs roles to approach using paths, stage at valid positions, clear angles, throw available utility, and reassess against delivered information. It does not simply set `round_win_chance += 10%`.

**Preference interactions — TARGET:** A strongly playbook-biased team with an IGL who performs well in rehearsed play should coordinate more multi-grenade lineups. A freeform or incompatible lineup can still throw utility but may mis-time it, spend it piecemeal, or miss. The grenades must be paid for and exist in inventory; cooperation never conjures free items.

## 14 — Match Modes, Rules & Tournament Identity

| Format | Rules and participation |
|---|---|
| **Standard Competitive** | MR12, up to 12 regulation rounds per half; first to 13 in regulation. Overtime policy OPEN. |
| **Quick Match** | MR8, up to 8 regulation rounds per half; first to 9 in regulation. Overtime policy OPEN. |
| **Real-time Queue** | Customers participate live against each other; server owns results and authorized coaching decisions. |
| **Live Tournament** | Organized customer competition with bracket/scheduling logic; formats and prizes OPEN. |
| **Opt-in Offline Engagement** | A customer explicitly allows their stored roster to participate while absent in eligible scrims and tournaments. |
| **Bot Scrim** | Practice against AI rosters; prototype treats results as nonpersistent for competitive history and chemistry. |
| **Bot Tournament** | Supported as a mode concept; reward/record eligibility OPEN. |

**Tournament organizer identity — LOCKED:** Real professional player and team identities are intended, but tournament organizers should be **original to CSTCG or merely implied** rather than directly reproducing real TO names and marks. Examples of fictionalized event presentation may be designed later; no final event-brand portfolio is approved.

**Manual coaching synchronization:** In live competition, both players must see the same round state. Pauses are server-authorized and replayable. Offline engagement uses Auto Coaching with an immutable lineup/playbook snapshot at match start. Fraud-sensitive match decisions cannot rely on local client clocks or files.

## 15 — Simulation Doctrine & Round State Machine

**Fundamental rule — LOCKED:** Play the round; don't pick a winner and invent a plausible narrative. Every significant action should change a real state: health, money, inventory, sightlines, known positions, access to cover, utility areas, bomb position, cooldowns, or remaining time.

**Round phases — TARGET:** BUY → SETUP → APPROACH / INFO → CONTEST → COMMIT / EXECUTE → SITE CLEAR → PLANT / RETAKE / SAVE → ROUND END. These are coordinated tactical intentions, not rigid sequences; early firefights, denied plants, lurk actions or late fakes can interrupt them. Every action requires a legal starting condition and produces observable effects.

**Simulation cadence — PROPOSED:** Movement/combat at a deterministic update rate; lower-frequency or event-triggered tactical planning; separate client rendering cadence. The authoritative engine logs structured events. Replay correctness requires stable ordering and seed handling under fixed game-patch data.

**Information asymmetry — LOCKED as a design property:** A team only knows what its players saw, heard or learned from teammates, with communication delay. The omniscient server knows all positions; a card AI must not automatically use this knowledge. Negative information (nothing seen on a watched lane) is evidence but not proof of an unoccupied area.

**Observable round outputs:** Kills, damage, ADR, entries, trades, assists, utility impact, rotations, space creation, economy, survival, objective actions, clutch scenarios, and the Impact Attribution ledger; not just final score.

## 16 — Transition to True 3D Dust II

The user-provided STL is the reference for 3D navigation and eye-level sightlines.

### 16.1 Decision and inputs — LOCKED / PROTOTYPE

**Spatial representation — LOCKED:** Detailed **tactical 2.5D initially** was selected for quick prototyping, but persistent geometry, elevation, wall, and teleport problems led to a subsequent explicit request to **move the simulation to true 3D**. The current intended architecture is therefore a **3D authoritative simulation** with an overhead tactical view, not a permanent two-layer sprite map.

**Uploaded assets:** A high-contrast top-down Dust II radar and a **`de_dust2+CS2+map.stl`** 3D geometry model. The radar is suitable for UI; the STL is a geometric source for world collision and navigation. The historical 2.5D prototype derived floors and sightlines from a 1024×1024 radar, but visual pixels do not reliably capture cover height, overlapping levels, or Source 2 wall materials.

**3D spatial prototype v1.4 — PROTOTYPE:** Converted the STL into renderable model geometry and static collision; baked an AStar3D surface-route graph; instantiated ten simple 3D agents; checked eye-level sightlines with physics rays; added overhead/angled cameras and an interactive Shift-click sightline probe. Offline inspection reported approximately 49,600 triangles, about 24.3 model units of height, ~4,073 navigation nodes, and a largest connected component of ~2,766 nodes. Those are **prototype model diagnostics**, not gameplay performance targets or verified playable map completeness.

**Crucial status distinction:** v1.4 is a **3D spatial test bench**, with the previous complete economy/utility/tactics/Impact Attribution simulator preserved in a separate **legacy 2.5D scene**. The full match system **has not yet been migrated** to the 3D world, and no claim should imply that a full competitive 3D match is currently integrated or runtime-proven.

### 16.2 Geometry and navigation contract — TARGET

Navigation must use the **actual connected walkable surfaces**, a player-sized agent radius and step-height constraint; stairs and ramps have real traversable geometry; overlapping plan-view coordinates do not connect floors. A direct path from CT Ramp to upper Catwalk is invalid unless the agent traverses legitimate connectors. Planar teleport fixes, force-snapping between disconnected floors and arbitrary vertical position changes are forbidden outside explicitly tested respawn/replay transitions.

**Map semantics to author:** T and CT spawn volumes (the user-defined green boxes); both bomb plant volumes; A Long, A Short, Cat corner, Mid Doors, Suicide, Upper/Lower Tunnels, B Doors, B Window, ramp, Goose and Car; legal flash lineups; common post-plant holds; penetrable materials; and smart cover points. A raw STL does not contain these game semantic labels.

**Collision and height:** Walkability is distinct from optical visibility and bullet penetration. Use 3D capsule/swept collision for movement, line traces from eye or muzzle for aim, and material-aware penetration for wallbangs. The overlay should not be used as an authoritative collision mask once 3D geometry is validated.

### 16.3 3D migration acceptance — TARGET

The same match engine should be able to swap its world interface to 3D geometry while preserving round rules, cards, economy and event attribution. Require: no stuck-in-Mid loops; no Mid-to-Cat position jumps; CT/T spawn correctness; legal A and B paths; accurate B wall/window distinctions; CT Ramp/Cat vertical isolation; credible Mid Doors/Suicide and Long sightlines; plausible post-plant cover, utility arcs, and actual collision; headless full-match deterministic replay.

## 17 — Agent Perception, Information & Control of Space

**Physical Control — LOCKED distinction:** Where teammates can actually occupy and traverse with relative safety, based on presence, approach geometry, cover and credible enemy threat. A card's nominal assignment alone does not guarantee this control.

**Information Control — LOCKED distinction:** What the roster **believes it knows** about enemy positions, utility, sounds, rotations and previously observed zones. Observations have time, confidence and source; reports spread after communication delay and can expire or be wrong. A revealed enemy does not remain permanently tracked through walls.

**Threat Control — LOCKED distinction:** The region an opposing team cannot safely enter without exposing itself to danger—held sniper angles, crossfires, suppressive fire, fire zones, un-cleared close angles, smoke gaps or known trade setups. A defender can deny a choke without firing or killing anybody.

**Implementation — TARGET:** The engine maintains a true spatial state and separate per-team belief maps. Threat derived from true knowledge is not automatically transferred to opponent knowledge. An attacker approaching Catwalk may pause because the corner is plausibly threatened, throw a smoke, stage a double-peek, or rotate while still uncommitted. It cannot confidently avoid an unseen defender through omniscience.

**Impact accounting:** A defensive smoke earns space-control credit only if it demonstrably changes an opponent's action or timing; mere throw count, presence, or intended utility is insufficient. See Impact Attribution below.

# 18 — Tactical AI: Teams Behave as Teams

## 18.1 Formation, entry and trade logic

Entry players lead; support players follow at **tradeable but staggered** spacing; the bomb carrier should not expose itself as the first body unnecessarily. Teammates should clear separate angles instead of standing in a single file. An actual entry death can cause an immediate trade attempt based on a recent credible location of the killer, not on server knowledge through the wall.

**Choke contests — TARGET:** Site takes have setup pauses before coordinated utility, bounded risk evaluation, and a meaningful 90-degree Cat corner contest. Defenders may hold an angle, delay with smoke/fire, bait utility or fall back. Attackers can stage, flash, double-peek, or choose another lane before a committed breach. Contest timing must be bounded so rounds do not stall indefinitely.

## 18.2 Site commitment and adaptive rotations

**Known historical bug:** Earlier prototypes often took B, walked back through Mid, then returned, without killing or clearing site. Normal route-selection and stalled-path recovery could overwrite the current execute. **Target solution:** A stateful commit lock after crossing a credible site-breach threshold. Once committed, normal agents must clear, secure, plant, defend, recover a lost bomb, or perform a real emergency withdrawal; generic route rerolls must not pull the team back to Mid.

**Not an absolute no-retreat law:** Real Counter-Strike teams sometimes abort attacks. A genuine bomb recovery, unmanageable utility, overwhelming defending presence, or extremely late strategic reversal can justify retreat if the game explicitly models the cause. Those exceptions must be rare enough to avoid repetitive oscillation.

## 18.3 Mid pressure and sniper setups

**TARGET:** More Mid Doors, Cat and Mid-to-B contact than in early prototypes; T AWP may hold Suicide/Mid Doors, CT AWP may hold Mid Doors; an upper Cat defender can contest the 90-degree pinch. AWP positions should be chosen by 3D sightline, cover and tactical intent, not artificially limited weapon range. Long angles are legal engagements when there is actual line of sight; hit probability and damage vary with distance.

## 18.4 Rotations, retakes, fakes and saves

CTs use communicated sightings and risk scores; one vague contact does not cause the entire defense to abandon the opposite site. Genuine fakes occupy or threaten an alternate lane and send plausible information rather than simply setting a `fake=true` flag. Retakes can coordinate multi-entrance timings, defensive utility and kits; a save

preserves real weapons and money while risking objective loss. An aggressive retake should be a deliberate, character-influenced choice, not a teleport to the bomb.

## 19 — Combat, LOS, Ballistics & Penetration

**3D LOS is the new authority — TARGET.** A clear eye-level/muzzle-level ray from an agent to a target can permit a firearm engagement at any distance on the authored map. There is **no artificial universal maximum firearm engagement distance**. Weapon effective-range data instead influences accuracy, damage falloff, recoil and willingness to take the shot. Knife and Zeus keep real melee/close-distance limitations.

**Realistic cover rules:** A wall blocks direct visibility; a ballistic trace may separately permit a reduced-accuracy, reduced-damage wallbang through authorized thin or penetrable materials **only when the attacker has credible knowledge or a justified spam angle**. Solid B-site partitions should not act like open windows. Open B Window and Mid Doors lanes must remain legitimate. Shooting across overlapping vertical floors requires a real unblocked 3D ray.

**Wallbang presentation — TARGET / PROTOTYPE:** Display ▥ alongside verified wallbang eliminations in the killfeed, with a distinct debug tracer. Store wallbang=true, penetrated materials, remaining power and event source for attribution and replay. Never award a wallbang because a projectile visibly passed through an incorrect radar mask.

**Damage/engagement pipeline — PROPOSED:** Visibility/perception → target selection → aim/facing → reaction → weapon cadence/ammo → accurate/hazard-weighted ray → material penetration → damage/armor/headshot → death → trade candidates → attribution. Use independent randomness only when appropriate, with deterministic random streams per match and patch.

## 20 — Complete Weapon, Equipment & Advanced Sniping Design

**Weapon coverage — TARGET / PROTOTYPE:** All **34 conventional CS2 firearms** plus Knife and Zeus x27 are represented in the v1.1 simulator catalog. Prices, side restrictions, damage, fire rates, magazines and kill rewards were authored as a simulation approximation; they must be reviewed against the current release data before launch.

| Category | Items in catalog |
|---|---|
| Pistols | Glock-18; USP-S; P2000; P250; Five-SeveN; Tec-9; CZ75-Auto; Dual Berettas; Desert Eagle; R8 Revolver. |
| SMGs | MAC-10; MP9; MP7; MP5-SD; UMP-45; P90; PP-Bizon. |
| Rifles | AK-47; M4A4; M4A1-S; Galil AR; FAMAS; AUG; SG 553. |
| Sniper rifles | AWP; SSG 08 (Scout); G3SG1; SCAR-20. |
| Heavy | Nova; XM1014; MAG-7; Sawed-Off; Negev; M249. |
| Other | Zeus x27; Knife. |

**Equipment slot model — OPEN:** Earlier versions use one active weapon slot; a production-ready implementation needs primary/secondary/melee, magazine/reserve ammunition, reload, dropping/picking up weapons, armor/helmet, defuse kit, and multiple utility slots. Weapon eligibility depends on team side and user loadout rules, which require final policy.

**Sniper specializations:** Passive, Active and Combat Sniping are visible only when inspecting a card, not on its collectible face. They modify *how* the agent holds angles, repositions, takes combat snaps, and reacts under pressure; they do not dictate unconditional hit chance.

## 21 — Grenades, Lineups & Reactive Utility

**Utility types — TARGET / PROTOTYPE:** Smokes, flashes, HE grenades, Terrorist Molotovs, and CT incendiaries. Grenades are physical and inventory-bound: they have origin, trajectory, time in flight, collision/detonation, area effects, expiration, and attribution.

**Flash presentation — TARGET:** Draw a curved flight arc from the thrower toward the flight endpoint; emit a white circular detonation pulse; mark blinded agents with a **spiky white outline** while blind. **Direct pre-detonation collision with an enemy causes exactly 1 HP contact damage**, stopping the grenade before its detonation sequence. Contact damage and confirmed blindness must be separately recorded.

**Flash reactions — TARGET:** A blinded player may retreat, remain behind cover, or spray randomly along a *previously held direction*, not snap to unseen targets. Blind spraying consumes ammo, respects firing rate, can hit allies, and collides with real walls. Turning away before detonation can reduce/avoid blind depending on vision geometry, but not every defender must face away by default.

**Fire reactions — TARGET:** An agent in damaging fire attempts to move to a legal safe location rather than stand in it. Molotov/incendiary zones temporarily deny physical traversal and pressure rotations. HE applies radius- and cover-sensitive damage. Smokes deny sightlines but are not universal opaque walls for movement or physical bullets.

**Lineup frequency and quality — LOCKED direction:** When the coaching slider favors **Playbook** *and* the IGL has good playbook compatibility, coordinated executes should become more likely to use **multiple smokes, stacked Molotovs, HE grenades, and flashes** in staged sequences. Freer teams may still use lineups. An individual with low Utility who rolls poorly can **miss** an authored lineup, producing an actual off-target landing position and tactical consequences.

**Constraint:** Utility availability is constrained by purchases, inventory, side restrictions, and time; team coordination never generates free grenades. The spectators' default feed should not be flooded by each grenade call; tactical details belong in optional play-by-play and impact inspection.

## 22 — Round Economy & Buy Decisions

**Economy — TARGET / PROTOTYPE:** Separate match dollars from account Credits. Each team assigns a buy policy (PISTOL / ECO / HALF / FORCE / FULL), while individual agents buy within their wallet and role priorities. Starting cash and payout table should follow the chosen CS2 rules version and be stored with the simulation for reproducibility.

| Current prototype economy assumption | Value / rule |
|---|---|
| Starting money per player | $800 at half start |
| Wallet cap | $16,000 |
| Loss-bonus steps | $1,400 / $1,900 / $2,400 / $2,900 / $3,400 |
| Ordinary round win | $3,250 |
| Defuse / explosion win | $3,500 |
| Individual plant / defuse action | $300 |
| Lost round after bomb planted and defused (T) | Additional $600 per T player in prototype |
| AWP kill | $100 in prototype |
| Typical rifle kill | $300 in prototype |
| Typical SMG kill | $600 in prototype |

**Status:** These are **prototype balance/rules assumptions**, not locked live-service constants. The prototype uses a loss-counter approach where a win decreases rather than automatically resets the counter. Audit against the applicable CS2 patch and user expectations before production lock.

**Round settlement:** Award the winner and losers according to the actual outcome, credit eligible elimination/plant/defuse actions exactly once, carry allowed saved gear to the next round, and reset cash/equipment at halftime. Armor, helmets and defuse kits have functional combat/objective effects. Economy should influence save vs force decisions and lineup availability.

**Future economic sophistication — OPEN:** Weapon drops and recovery, weapon loadout choices, team drops/shared purchasing, full item pricing, overtime bank, kill reward variants and team-save modeling.

## 23 — Bombsites, Plant Choices, Post-Plant & Retake AI

**Plant requirement — TARGET:** Entering the bombsite center is insufficient. Attackers should establish real physical control, check relevant angles, account for live defenders, and use an actual plant location. A site can be secured without a kill if the available space is legitimately safe enough.

**Unconventional plants — TARGET:** Allow multiple validated sites within A and B, including positions chosen for cover, escape route, utility and post-plant lines. The bomb actor and UI both use the true plant coordinates; a CT must physically reach those coordinates to defuse. No decorative marker at a default plant center should override the actual bomb position.

| B anti-retake hold | Intended coverage |
|---|---|
| **B Car** | B Doors entrance and closer retake routes |
| **B Deep Plat** | Deep cover, crossfire and tunnel-side defense |
| **B Window** | Window retake threat |
| **B Tunnel Flank** | Backstab through tunnels |
| **B Doors Crossfire** | A second angle on defenders leaving Doors |

| A anti-retake hold | Intended coverage |
|---|---|
| **A Car** | Short/A Ramp angles and nearer defensive lanes |
| **Goose** | Deep A cover and controlled retake engagement |
| **A Long Ramp sniper** | Longer-range Long approach on correct actual elevation |
| **A Short anti-retake** | Catwalk access |
| **A Default / Ramp crossfire** | Plant region and ramp approaches |

**Post-plant decisions — TARGET:** Assign meaningful holds based on weapon, role, bomb position and known enemy approaches, but then **adapt dynamically** as information changes. An AWPer on Long should reconsider the position if a coordinated Short retake becomes likely. The retaking CT team should synchronize utility and entry lanes instead of each rushing the C4 independently.

## 24 — Spectator, Presentation & Accessibility

**Primary spectator — LOCKED: Top-down tactical presentation plus killfeed**, not a fully 3D FPS broadcast. The authoritative world can be 3D while the customer sees a readable radar. A temporary angled 3D developer camera and sightline/debug overlays are for engineering, not mandatory customer interface.

**Default feed — LOCKED:** Only **eliminations, bomb plants and successful defuses** appear in the primary feed. **Play-by-play is optional/toggleable**, revealing rotations, utility, staging, reads, strategy choices, misses, saves and other tactical events. Wallbang kills display the chosen wallbang symbol. Maintain clear color-independent labels for accessibility.

**Utility visualization — TARGET:** Smokes as temporary sight-obstruction volumes; orange fire zones for Molotov/incendiary hazards; curved flash arcs, white detonation ring, and white spiky blind outline; HE

trajectories/explosion; kill tracers and wallbang effects. In 3D, utility effects originate from real trajectories and are projected to the radar.

**Player inspection:** Select an agent to see role, team, HP, money, current weapon and gear, active blind/fire status, current intent, observed tactical behavior, relevant advanced stats (in card views), and actual input to match decisions where appropriate. Do not expose the hidden archetype.

**Interface views — TARGET:** Onboarding; Collection; Card Detail; Training and Mods; Team Builder; T/CT Playbook; Matchmaking; Live Spectator; Postmatch Scoreboard; Impact Events; Performance Rating; Marketplace; Packs/Store; Events/Challenges; Team Chemistry Journal; Settings. Mobile groups inspector panels into collapsible drawers.

## 25 — Impact Attribution & Performance Rating

**Impact Attribution — PROTOTYPE integrated in the legacy 2.5D sim:** Record real causes of effective damage, kills, utility blindness, verified trade support, saved equipment, planted/defused objectives, meaningful clutch contributions and enemy decisions changed by created space. Each event has an ID, round/time, actor-card ID, details, supporting awards and reproducible explanation.

**One elimination, one shared budget.** A killer, flash helper, trader and space creator can share a single kill-event allocation. Do not stack several full kill credits for the same death. Damage credit is capped at target HP actually removed, preventing overkill inflation; a blind earns verified flash credit only if an enemy was actually blinded.

**Positioning credit requires influence.** Standing in an arbitrary zone or throwing a smoke does not automatically earn Impact. If the utility and positioning actually delays a rotation, forces an approach change or prevents safe entry, the resulting causal event may receive credit.

**Performance Rating — PROTOTYPE:** Separate from permanent card statistics and a raw OVR. Current provisional categories/weights: Damage **20%**, Eliminations **16%**, Utility **18%**, Trading **17%**, Survival **10%**, Clutch **9%**, Positioning/Objectives **10%**. The prototype normalizes relative to opportunities and shrinks rare events toward baseline. **Weights and expected performance baselines are experimental, not validated professional rankings**.

**Archive scope — PROTOTYPE:** Per exact card ID, with manager/team identity snapshots; rolling 20-match results and a session-level competitive history. Current prototype **scrims are ephemeral**: they produce inspection logs but no persistent career stats, chemistry, rewards or progression. Treat these exclusions as prototype rules to confirm before launch, especially for bot tournaments.

## 26 — Pack Ecosystem, Prices & Drop Pools

**Standard map packs — LOCKED:** Each ordinary map pack contains **exactly 1 player card + 2 modification cards**. Players can appear in several named map-pool collections with different weights; no exclusive identity rule. Examples discussed: **Dust, Mirage, Cache, Nuke, Inferno** and additional future map collections.

**Launch store mix — LOCKED: Seven default pack offerings: 3 super-cheap, 3 mid-range, 1 expensive**. Exact pack names, prices and pool definitions OPEN. Ordinary player rarity odds are pack-specific; Peak lies within the earlier 0.50%–0.001% range where applicable. No numerical Divine odds are locked.

**Sealed packs — LOCKED:** A purchased pack is an actual transferable item, not automatically opened. Its original edition, source collection, published odds and generated/linked drop-table version must be auditable. Older sealed event packs remain tradable after the event; opening them later preserves the original edition and its documented collection eligibility.

**Recommended supply implementation:** Publish odds and pool versions before release, protect opening logic server-side, do not silently mutate an already minted sealed pack's eligibility table, prevent duplicate item

creation, and preserve an audit trail of every pack-to-card mint transaction. This is a technical recommendation subject to legal/platform requirements.

## 27 — Events, Artifact Packs, Shiny & Provenance

**Events — LOCKED concept:** Time-limited tournament, era, history, or commemorative collections; event challenges that provide free limited cards or packs; tradable sealed event packs; some **Shiny Event Packs**; anniversary reissues or returning pools so gameplay content is eventually available again. Original editions retain distinct visuals/provenance even if reissued content is statistically equivalent.

**Shiny Event guarantee — LOCKED:** At least **one Rare-or-better card** within the pack, and better rarity odds for the others. The exact normal content count of all event-pack variants and precise adjusted distributions remain OPEN; do not quietly assume every event pack matches standard 1+2 composition without an explicit definition.

**Artifact Pack — LOCKED:** An expensive **single collectible** draw from **inactive/discontinued pack collections**, with **no rarity floor**. Choose **one inactive source pack at random first**, then sample **one eligible item** from that pack's table. Eligible types can include a player, modification, eligible tradable strategy or future approved collectible. Common/Floor outputs remain possible and potentially frequent.

**Shiny Artifact Pack — LOCKED:** One item, **guaranteed Rare or higher**. A Shiny Artifact can draw from relevant inactive source pools under its specific published and versioned distribution. The guarantee must apply to the chosen item. Exact Rare/Legendary/Peak/Divine probabilities are OPEN.

**Price relationships — LOCKED:** Let $P$ be the price of the most expensive ordinary/default pack. **Ordinary Artifact** = **1.10 × P**. **Shiny Artifact** = **1.45 × Ordinary Artifact** = **1.595 × P**. This supersedes older wording that incorrectly compared Shiny with an ordinary default pack.

**Inactive eligibility delay — TENTATIVE:** Approximately **six weeks** after retirement before a collection becomes Artifact eligible. The six-week period was suggested, not irrevocably approved.

**Provenance — LOCKED:** Record original collection (e.g. Cache Pack S1), edition, acquisition source, item ID and opening origin. Items obtained from Artifact draws display an **"A" provenance symbol**, even if their actual play attributes match an earlier original-edition card. Opening an *original sealed event pack later* does **not** add an Artifact A stamp. Distinguish source, edition, rarity and cosmetic Shiny appearance.

## 28 — Marketplace, Trading, Credits & Monetization

**Marketplace functionality — LOCKED:** Customers can trade/sell eligible player cards (including fully modified or promoted instances), mods, approved strategy cards and unopened packs for Credits. Trading formats: **fixed-price listings, auctions, and direct peer-to-peer trades**. A marketplace transaction fee **does apply**, although the exact rate is OPEN. A previous 5% suggestion is not a locked number.

**Card disclosure:** Show professional/character identity, rarity, tier and slots, base and modified stats separately, cooldown state, edition/provenance, discovered observational history, and eligibility. Never display literal Alpha/Bravo/Charlie values. Seller-generated descriptions and price history should be separated from server-verified properties.

**Free-to-play — LOCKED:** Credits are earned in the game and may be bought with real money. This creates material pay-to-progress and consumer-law risks, especially when a tradable random pack is purchased with bought currency. Before production monetization, review regional rules on paid random items, odds, minors, payment reversals, refunds, chargebacks, item resale, tournament prizes, and fraud. No real-cash redemption/cash-out system has been approved.

**Technical integrity** — **TARGET:** Transactional listing locks and idempotent credit transfers, a double-entry or reconciled credit ledger, immutable transfer history, auction escrow, P2P trade confirmations, abuse throttling, pricing disclosures and server-side ownership checks.

## 29 — Server, Multiplayer & Persistence Architecture

**Engine direction** — **LOCKED provisional choice: Godot 4.x** for PC/mobile clients. The recommended architecture is to separate the eventual competitive core into a headless, server-authoritative simulation (C#/.NET is the proposed backend language), while Godot visualizes accepted events. Existing experimental simulation scripts are currently mostly GDScript; migrating them to .NET is an architectural recommendation, not an achieved step.

**Recommended service layers** — **PROPOSED:** Game client; authentication/profile; inventory/catalog/pack opening; marketplace/credit ledger; queues and tournaments; match orchestration; deterministic simulation workers; historical results/Impact Attribution; content version service; customer notifications. PostgreSQL for persistent transactional records and Redis for caches/queues are candidates, not locked contracts.

**Online trust boundary:** Clients submit permitted choices and inspect outcomes. They **never** choose winners, generate paid packs locally, alter card rolls or post arbitrary rewards. Matches use locked roster snapshots, content data versions, legal coaching calls and a deterministic input/event log. Unattended offline engagement must be explicit opt-in with a server-authoritative plan.

**Performance considerations:** Keep simulation ticks and AI decision cadence separate from rendering; avoid per-frame omniscient tactical searches; profile thousands of headless matches; batch bot tournaments; bound utility path searches and collision rays; store only necessary event detail with replay snapshots or logs.

## 30 — Data Model & Event Contracts

| Entity | Authoritative properties |
|---|---|
| ProfessionalTemplate | Identity, real-data source/copyright status, career/event template, role profile, per-stat ceilings. |
| PlayerCardInstance | UUID, professional or Pubstomper character ID, rarity, T-level, immutable rolled stats, hidden behavior, sniper profile, provenance. |
| AppliedModification | Card instance, mod definition, consumed/equipped state, slot tier, effective-stat delta, cooldown. |
| TrainingEffect | Card instance, stat delta, matches remaining, UTC expiration. |
| LineupSnapshot | Five unique identity IDs, exact card IDs, side roles, playbook, coaching slider, chemistry state. |
| ChemistryRecord | Current value, hidden ceiling, historic roster key, active depreciation stage, baseline and update events. |
| PackDefinition / SealedPack | Eligible pool, published odds, edition, source eligibility, guaranteed drops, provenance version. |
| MatchState | Mode/rules, two lineups, map/world version, seed, economic inventory, combat/utility/round state. |
| MatchEvent | Stable event ID, patch and timestamp in sim, actor, world positions, kind, outcome and attribution. |
| MarketplaceListing | Seller, item lock, ask/auction state, fee rule, escrow and audit log. |
| CreditLedger | Server-verified debits/credits, transaction reason, reconciliation and refunds. |

**Event example (illustrative):** A planted-bomb retake could emit flash_throw, flash_bounce, flash_detonation, blind_applied, info_delivered, crossfire_peek, damage, kill, trade_window, bomb_defused, round_end, each with authoritative event time, agent IDs and world position. Events should not retroactively grant credit for interactions that didn't occur.

## 31 — QA, Telemetry & Balance Acceptance

**Primary development philosophy:** Resolve game-state correctness, geometry and repeatability *before* introducing more collection systems. Historical prototypes repeatedly exposed typed-array errors, duplicate dictionary keys, invalid Mid lookups, stuck movement, teleport-like floor jumps, overly permissive B walls, aim direction issues, and unnecessary B-site exits. These issues justify a mandatory automated Godot parser/runtime gate for every release, not static Python checks alone.

**Required tests — TARGET:** Headless Godot editor parse; scene import; complete MR8/MR12 simulations; navigation routes for every spawn/site/rotational lane; no unauthorized teleport or boundary crossing; line-of-sight ray results for known angles; wallbang material correctness; no friendly information leaks; legal purchases; direct-flash 1 HP contact; stun/fire response; multi-grenade lineups with successful/missed targets; no routine abandonment after commit; planting only in legal and credibly controlled zones; coherent post-plant positions; attribution invariants; replay-seed determinism; scrim persistence isolation; five-distinct-identity roster validation.

**Batch telemetry — TARGET:** Distribution of kills, assists, ADR, openings and AWP engagements; average rounds; percentage of rounds with a Mid fight/Cat contest; execute setup time; fraction of committed-site reversals; CT overrotation frequency; percentage of stuck agents; number of unreachable path targets; proportion of lineups landing correctly by Utility stat and coaching fit; economy buy/save patterns; accessible win rates for lower-rarity/high-chemistry teams.

**Balance hypotheses to test:** Strong Aim should not erase disadvantages of bad position or poor equipment; excellent team play should yield measurable advantages without flat 'chemistry increases Aim'; unusual high-rarity profiles should remain situationally viable; alternative T strategies should produce distinct outcomes; a seeded match with the same inputs should match replay traces exactly.

## 32 — Development Status Through v1.4

| Prototype milestone | What is present or attempted | What is **not** proven / done |
|---|---|---|
| v0.1–0.4 | Ten-agent 2D match, basic economy, routes, Dust II radar, elevation masks. | Accurate geometry and reliable Godot runtime across releases. |
| v0.5–0.8 | Impact Attribution, chemistry session state, site takes, Cat/Mid changes, play-by-play toggle. | Robust pathfinding and production event accounting. |
| v0.9–v1.1 | Choke risk, grenade effects, economy, weapon catalog, sniper profiles, control maps, lineups. | Calibrated professional behavior; complete inventory and true 3D collision. |
| v1.2 | Fire/flash responses, blind spray, named anti-retake positions. | Dynamic repositioning and consistently correct cover. |
| v1.3 | Site lock refinement, unlimited firearm engagement range, B wallbang/radar-derived ballistics, Mid AWP angles. | Pixel-perfect Source 2 geometry; all-angle 3D occlusion. |
| **v1.4** | Imported STL, 3D collider and AStar routes, ten test agents, raycasts, 3D angle tool. | **Full 3D match integration.** Legacy full-match scene preserved separately. |

**Validation caveat:** Prior project deliveries report extensive **offline structural/geometric checks**, but the authoring environment generally lacked a Godot executable. Bugs identified through customer runtime testing demonstrate that prior 'passed static tests' are **not equivalent to end-to-end engine verification**. The team should not represent the combined match/3D project as released or production-ready.

## 33 — Roadmap, Dependencies & Exit Criteria

| Milestone | Exit criterion |
|---|---|
| **M0 — GDD consolidation** | This master document; decision register; validated rule ownership and open items. |
| **M1 — Geometry foundation** | Reliable Dust II 3D collision, walkable navigation, spawns, safe angle probes and coherent height traversal. |
| **M2 — Complete 3D match** | MR8/MR12, full 10-agent economy/combat/objective loop on the 3D world, headless reproducible tests. |
| **M3 — Tactics & telemetry** | Credible entry spacing, contest/commit, information, utility, retakes, saves, measurable improvement from stats. |
| **M4 — Player cards** | Professional + Pubstomper templates, unique rolls, rarity, hidden behavior, sniping, tiers/mods/training. |
| **M5 — Chemistry & review** | Correct growth/substitution curves, observed interactions, Impact Attribution, performance history. |
| **M6 — Packs & store sandbox** | Seven default packs, starter gift, Artifact/Shiny rules, provenance and verified odds; no paid purchases in testing. |
| **M7 — Secure multiplayer** | Authoritative queues, opt-in offline simulation, coaching timeouts, tournament orchestration, persistent accounts. |
| **M8 — Market and economy** | Fixed/auction/P2P exchange, fee, credit ledger, fraud mitigation and item lock. |
| **M9 — Release readiness** | Rights review, paid random-item compliance, telemetry-based balancing, mobile performance and accessibility QA. |

**Recommended immediate next work:** Build a strict geometry adapter between the full match state and STL-driven Godot 3D world; run a fully seeded 10-player round and then MR8 through that adapter; only after the movement, LOS and bomb-loop regressions are clean should advanced tactical AI be copied onto it.

## 34 — Risk Register & Safeguards

| Risk | Why it matters | Mitigation direction |
|---|---|---|
| 3D geometry is inaccurate/nonwatertight | Hidden holes, unsupported surfaces, bad player collision. | Mesh QA, walkability baking, authored zone tags, test ray corpus. |
| AI repeatedly gets stuck or teleports | Ruins realism and match outcomes. | Real obstacle clearance, failure recovery, maximum displacement assertions. |
| Automated teams use hidden enemy truth | Breaks information gameplay. | Separate belief stores, delayed messages, authoritative trace logs. |
| Statistics overpower tactics | Expensive cards dominate irrespective of decisions. | Sim scenarios, tradeoffs, model sensitivity and high-volume calibration. |
| Chemistry feels arbitrary | Player loses agency to invisible systems. | Event-linked journal, trends, consistent hidden ceiling rules. |
| Paid credits + randomized tradable packs | Regulatory, platform and fairness risks. | Jurisdictional legal review, odds disclosure, minors/limits policy, abuse prevention. |

| Risk | Why it matters | Mitigation direction |
|---|---|---|
| Real pro/team/Valve rights assumed | Commercial distribution may be blocked. | Written license/data rights review and fallback original content. |
| Auction/P2P fraud | Economy trust loss. | Immutable ledger, escrow, atomic trades, audits, monitoring. |
| False claims about build readiness | Wasted development cycles. | Mandatory Godot parser and runtime regression gates before 'tested' claims. |
| Artifact supply dilution/price confusion | Collectible value and trust erosion. | Versioned pool/editions, correct prices, transparent reissue rules. |

# 35 — Decision Register: Locked vs. Still Open

## 35.1 Confirmed numerical & product rules

| Decision | Authoritative current value |
|---|---|
| Platforms | PC and mobile |
| Match opponents | Primarily PvP; bot modes; opt-in offline match engagement |
| Monetization | Free-to-play; earnable/purchasable Credits |
| Starting five | 4 named Pubstompers + choice of 1 of 20 Common T5 pros |
| Starter gift | 1 guaranteed Peak player + 4 additional generated cards |
| Rarity | Floor, Common, Uncommon, Semi-Rare, Rare, Legendary, Peak, **Divine** |
| Promotion | T5→T4→T3→T2→T1; 10/25/40/55 points |
| PP by rarity | Floor 1, Common 1, Uncommon 2, Semi-Rare 3, Rare 3, Legendary 4, Peak 5, Divine 6 |
| Slots | 2 new tier slots per promotion; 10 at T1 |
| Pubstomper progression | 10× standard promotion requirement |
| Modification cooldown | 10 matches or 10 hours, whichever comes first |
| Training duration | 3 matches or 2 hours, whichever comes first |
| Standard pack | 1 player + 2 mods |
| Default pack store mix | 3 cheap + 3 mid + 1 expensive |
| Peak odds | 0.50% to 0.001% depending on pack |
| Artifact / Shiny | 1 item from inactive source pool; normal no floor; Shiny Rare+ |
| Artifact price | 110% of highest-priced standard pack |
| Shiny Artifact price | 145% of regular Artifact pack |
| Artifact eligibility | ~6-week delay suggested, not locked |
| Event Shiny | ≥1 Rare+ and better odds for remaining cards |
| Market modes | Fixed price, auctions, P2P; transaction fee applies |
| Chemistry | Starts at 0; hidden composition-only ceiling |
| Roster substitution | 1: $-10 \rightarrow C_0 - 30$; 2: $-10 \rightarrow C_0 - 40$; 3: $-15 \rightarrow C_0 - 40$; 4+: zero |
| Depreciation | 3 matches, then trend toward new hidden ceiling |
| Competition | MR12 standard, MR8 quick |
| Coaching | Auto default; manual optional; slider playbook vs. midrounding |
| Timeouts/calls | MR12 three tac pauses + Round 1/13 calls; MR8 one pause, no half call |

| Decision | Authoritative current value |
|---|---|
| Spectator | Top-down radar, killfeed + plant/defuse default, detailed play-by-play toggle |
| Organizer identities | Original/implied tournament organizers |
| Simulation geometry | True 3D world target; top-down spectator |

## 35.2 Outstanding decisions to resolve

7. Final commercial name, brand system and whether CSTCG remains the public-facing title.
8. Licensing scope and agreements for professional likenesses, team marks, Valve maps/weapons, and historical data sources.
9. Whether the starter gift's four additional cards follow player/mod/strategy categories, and exact starter-only pack pool.
10. Pubstomper rarity rules, promotion-card conversion math, trading, rename ownership and cap on created characters.
11. Individual Divine rarity odds and complete published odds for all 7 standard packs and event/Artifact packs.
12. Mod stacking, old-mod fate when replaced, cooldown state upon sale, and which games reduce match counters.
13. Training stacking, bonuses, costs, and bot/scrim eligibility.
14. Chemistry progression coefficient, negative lower bound, historic-roster volatility, 4+-replacement recovery and eligible mode rules.
15. Score/tiebreak overtime policy for MR8, MR12 and tournament play; timeout lengths and restrictions.
16. Progression/rank track definition; credit/XP rates; strategy unlock placement.
17. Detailed Pubstomper/real-player base stat formulas, rarity ceilings, professional season templates and card-inspector UI.
18. Exact asset ownership rules for auction escrow and direct trades; fee amount, listing duration and anti-inflation sinks.
19. Shiny Event pack composition, cosmetic and provenance differences; artifact sampling weights/eligibility exceptions.
20. Launch roster/map scope, full 3D Dust II geometry accuracy, additional maps and performance budget.
21. Full primary/secondary/inventory rules, utility carry/limits, weapon drops, loadouts and current economy version.
22. Match event eligibility for rewards, scrim history, offline tournament recording, bans and disconnection behavior.
23. Competitive ranking, matchmaking skill rating, seasons, anti-smurf and bot eligibility.
24. Which coaching decisions are legal during authorized tactical pauses and whether they consume in-game time.
25. Final Impact Attribution rating weights and contextual expected-value calibration.
26. Server backend choice, persistence schema, regional operations, paid currency compliance, analytics privacy and accessibility requirements.

## Appendix A — Sample End-to-End Match Story

A customer enters a live MR12 queue with four developed professional cards and one beloved Pubstomper. The team has chemistry 71 with a concealed upper limit; a new AWPer was introduced two games ago and the roster is still in its three-game depreciation period. The customer selects a Playbook-heavy approach, relying on an IGL who handles rehearsed utility well, and the team buys within its actual economy.

On Dust II the attackers show three players near Mid while the entry pair secures the B Tunnels approach. A CT AWPer controls Mid Doors from a credible long-range sightline and communicates an enemy sighting. That report updates the defenders' **Information Control**, but it does not reveal the rest of the attacking roster. A defensive smoke increases temporary **Threat Control** at Mid and delays one opponent without creating a kill.

The attackers commit B after staging, throwing purchased smokes toward Window and Doors, a flash for the entry and a Molotov toward a defensive hold. The first support flash is accurate; the second player's weak Utility roll causes a lineup miss. The entry clears the first angle and dies; the teammate behind it can immediately trade from a legal position. The bomb carrier waits until enough cover is secured to plant at a nondefault position.

Surviving attackers take B Car, Deep Plat and tunnel-side crossfire positions. A blinded CT retreats behind cover; another fires a desperate inaccurate spray without snapping to hidden enemies. The retake develops from Window and Doors, with the defuse-kit carrier moving to the **actual** bomb coordinates. The defending team's late-round decision depends on remaining time and money. Every inflicted HP, verified blind, trade, attempted plant and objective outcome appears in the event ledger; the default spectator sees only kills, plant and defuse, while play-by-play remains optional.

This story illustrates **desired causal capability**—not a claim that every described element already runs together in the current v1.4 3D prototype.

## Appendix B — Engineering Contracts & Minimal Test Matrix

| Test | Expected invariant |
|---|---|
| Same seed + same patch + same choices | Identical authoritative match results and ordered events. |
| Duplicated pro in one lineup | Validation fails, even if cards have different rarities. |
| T5→T4 promotion | Consumes 10 PP and leaves base stats unchanged; adds 2 T4 slots. |
| Higher-tier mod into lower slot | Invalid; lower-tier into higher slot allowed. |
| Training/modify expiry | First of qualifying match-count or wall-clock expiration triggers. |
| Artifact Pack | Random inactive collection first; one eligible item; no minimum rarity. |
| Shiny Artifact | One Rare+ item; correct relative pricing. |
| Four lineup substitutions | Chemistry reset to 0, with later progression rules defined explicitly. |
| No LOS to hidden enemy | No direct aimed shot unless valid wallbang knowledge/penetration. |
| Cat upper and lower overlap | Same (x,y) does not imply free movement or fire between levels. |
| Fire zone | Agent can attempt a legal safe escape and still take damage if trapped. |
| Flash impact | Exactly 1 HP contact damage before later detonation. |
| Missed lineup | Actual detonation position differs from nominal template target. |
| Breached site | Default planning cannot repeatedly send the whole team back to Mid. |
| Plant marker | Reflects true physical bomb coordinates; kit defuse time honored. |
| Kill impact ledger | Shares sum to one allocation, no duplicate killer event. |
| Default event feed | Shows kills, completed plant and defuse only. |
| Scrim history in current prototype | Never writes competitive history or permanent chemistry. |
| Offline opt-in | Roster is not matched without explicit customer authorization. |

## Appendix C — Glossary

**Attribute:** Immutable rolled ability score on a card, before mods and training. **Card instance:** Uniquely owned collectible with its own stats and history. **Rarity:** Distribution/ceiling class (Floor to Divine), not promotion tier. **Tier:** T5 to T1 progression controlling modification slots. **PP:** Promotion points received by sacrificing same-professional duplicates. **Pubstomper:** A customizable faceless original player card type. **Artifact A:** Provenance mark for a card acquired via an Artifact draw. **Shiny:** Special pack class with rarity guarantees, independent of edition origin. **Alpha/Bravo/Charlie:** Hidden, permanent simulation-only interpersonal dispositions. **Chemistry ceiling:** Maximum long-run chemistry allowed by hidden composition. **Playbook slider:** Degree to which the team follows prepared strategies versus improvising. **Space Control:** Independent Physical, Information and Threat assessments. **Commit lock:** Tactical constraint preventing arbitrary site abandon after a genuine breach. **Impact Attribution:** Event-based allocation of actual player contributions. **2.5D:** Map/radar with authored floor layers and approximated vertical relationships. **3D spatial test bench:** STL-derived world collision/navigation prototype that has not yet absorbed the full match engine. **TO:** Tournament organizer; the game's TO branding is original or implied.

**End of consolidated Master Game Design Document v2.0. Priority:** Integrate the existing full tactical match simulation into the validated 3D Dust II world, and build automated engine-level tests before enlarging the feature set.